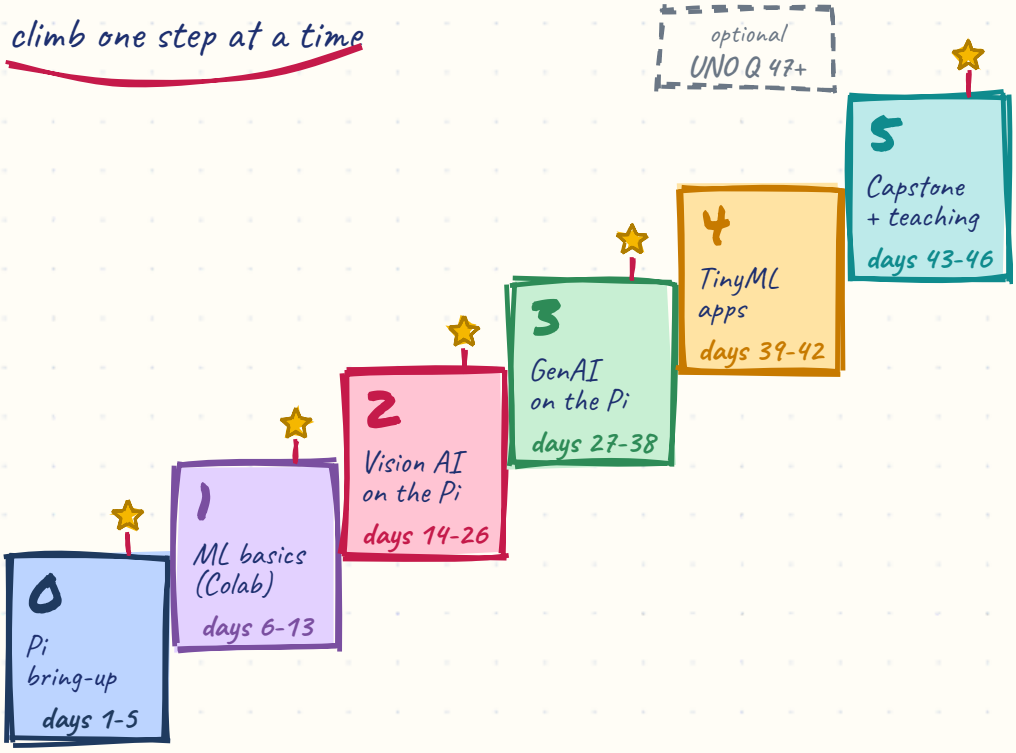
MY 46-DAY ROADMAP

Pi 4 (4 GB) / 2 Pis + 2 laptops / 2 hours a day / ~9 weeks

climb one step at a time



stars = what I can show: Pi ready (5) - first CNN (13) - own classifier on the Pi (18) YOLO counting (26) - RAG (32) - SLM drives GPIO (38) - capstone (46)

TWO PIS, TWO LAPTOPS: DO EVERY DAY TWICE

Pass 1 - learn
on Pi A, ~2 hours: tutorial + animation + lab + notes

Pass 2 - consolidate
on Pi B, from my notes only. Faster. Finds the hidden mistakes

- different host names: `raspi-office` / `raspi-home`, same user
- code + notes in GitHub, then `git pull` everywhere
- Phase 1 needs no Pi (Colab) - good for travelling

GET READY (BY DAY)

- ☐ **1** Pi 4, microSD 32 GB+, 5 V / 3 A supply, card reader
- ☐ **4** Pi camera (CSI) or USB webcam
- ☐ **6** Google account (Colab) **13** Edge Impulse account
- ☐ **35** USB mic + speaker
- ☐ **36** breadboard, wires, LEDs + resistors, button, DHT + BMP280 sensors

Honest limits: 4 GB = small language models only, slower than the book's Pi 5. Check Python vs TFLite on day 4-5. Day splits are estimates.

MY NOTES

my start date

which Pi first

DAY BY DAY

C = class number in the course repos (IESTI05 Edge AI, IESTI01 TinyML) · pp. = PDF pages of the book (printed + 1)

P0 · Pi bring-up *days 1-5*

1. Meet the Pi 4, flash the microSD, first SSH
   IESTI05 C3 · pp. 20-21, 25-27, 29-39
2. Remote access: file transfer, remote desktop, two Pis and two networks
   IESTI05 C3 · pp. 38-44, 56-61
3. Linux command line
   IESTI05 C2 (slides + linux_intro.pdf)
4. Camera, swap memory, updates
   IESTI05 C3 slide 10 · pp. 43-55, 62-64
5. Python virtual environment + Jupyter on the Pi
   IESTI05 C4 · pp. 65-75

P1 · ML foundations (Colab) *days 6-13*

6. Colab and Python review; what is TinyML
   IESTI01 C2-C4b
7. The machine-learning paradigm
   IESTI01 C5
8. Neural-network building blocks
   IESTI01 C6
9. Regression with a DNN
   IESTI01 C7
10. Classification with a DNN
    IESTI01 C8
11. Datasets, splits and metrics
    IESTI01 C9
12. Convolutions; image classification with a CNN
    IESTI01 C10-C11
13. Edge Impulse intro (Cifar-10); preventing overfitting
    IESTI01 C12-C13

P2 · Fixed-function AI on the Pi *days 14-26*

14. TensorFlow Lite + MobileNetV2 inference
    IESTI05 C5 · pp. 75-86
15. Live camera classification; a model trained from zero
    IESTI05 C5 · pp. 86-93
16. Custom classifier: goal and data collection
    IESTI05 C6 · pp. 94-102
17. Train in Edge Impulse
    IESTI05 C7 · pp. 103-111
18. Test, deploy and run live on the Pi
    IESTI05 C7 · pp. 112-130
19. Object detection fundamentals
    IESTI05 C8 · pp. 131-145
20. Pre-trained detectors and live stream
    IESTI05 C8 · pp. 146-153
21. Custom detection: data and labeling
    IESTI05 C9 · pp. 154-168
22. SSD MobileNet in Edge Impulse; deploy
    IESTI05 C10 · pp. 169-187
23. FOMO (lightweight detection)
    IESTI05 C10 · pp. 188-202
24. YOLO with Ultralytics on the Pi
    IESTI05 C11 · pp. 203-219
25. YOLO: custom training and deployment
    IESTI05 C12 · pp. 220-237
26. Counting objects (bees) + Part 1 mini-project
    IESTI05 C13-C14, Part 1 assignment · pp. 238-264

P3 · Generative AI on the Pi *days 27-38*

27. Text generation with an RNN (VerneBot)
    IESTI05 C15 · pp. 265-289
28. Knowledge distillation in practice
    book chapter · pp. 290-308
29. Small language models: first model with Ollama
    IESTI05 C16-C17 · pp. 309-335
30. Ollama with Python
    IESTI05 C18 · pp. 336-360
31. Function (tool) calling
    IESTI05 C19 · pp. 361-375
32. Retrieval-augmented generation (RAG)
    IESTI05 C20 · pp. 376-388
33. Agents with small models
    IESTI05 C20a · pp. 560 onward
34. Vision-language models (Florence-2)
    book chapter · pp. 389-433
35. Audio and vision AI pipeline
    book chapter · pp. 434-458
36. GPIO and physical computing: LEDs, buttons
    IESTI05 C21 · pp. 459-485
37. Sensors and notebooks
    IESTI05 C21 · pp. 486-510
38. SLMs for IoT control
    IESTI05 C22 · pp. 511-557

P4 · TinyML applications *days 39-42*

39. Anomaly detection and K-means
    IESTI01 C21-C22
40. Keyword spotting (concepts)
    IESTI01 C23-C24
41. Image apps on microcontrollers; data collection
    IESTI01 C25-C27
42. Responsible AI; going further with Edge AI
    IESTI01 C28-C29

P5 · Capstone + teaching plan *days 43-46*

43. Choose and plan the project
    IESTI05 Part 1 / Part 2 assignments
44. Build
45. Build, test and measure
46. Present, write up, and plan how to teach it

P6 · Optional: Arduino UNO Q *days 47-56*

47. UNO Q: foundations and setup
    IESTI05 UNO Q Ch0-Ch1
50. SLMs on the UNO Q; benchmarking; Python
    IESTI05 UNO Q Ch2 (C19-C21)
53. Multimodal and agentic AI
    IESTI05 UNO Q Ch3, Ch5
55. Physical AI: Bridge between the two processors
    IESTI05 UNO Q Ch4